

1. Graph of the following functions, labeling at least three major points on each sketch.

$$\begin{aligned} \text{a) } y &= -2\left(\frac{1}{2}x + \frac{3}{2}\right)^2 + 2 \\ &= -2\left[\frac{1}{2}(x+3)\right]^2 + 2 \end{aligned}$$

$$(x, y) \rightarrow (2x-3, -2y+2)$$

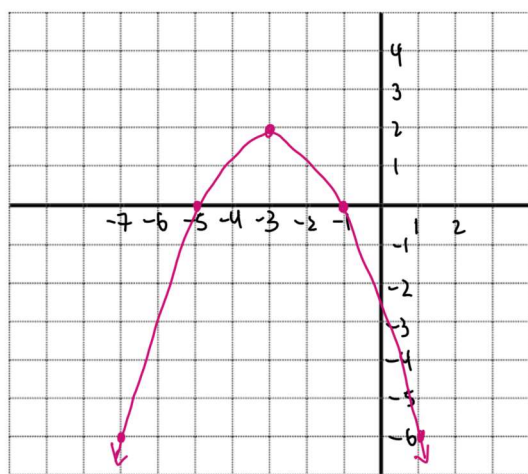
$$(-2, 4) \rightarrow (-7, -6)$$

$$(-1, 1) \rightarrow (-5, 0)$$

$$(0, 0) \rightarrow (-3, 2)$$

$$(1, 1) \rightarrow (-1, 0)$$

$$(2, 4) \rightarrow (1, -6)$$



$$\begin{aligned} \text{b) } y &= \frac{1}{\frac{1}{2}x+2} - 2 \\ &= \frac{1}{\frac{1}{2}(x+4)} - 2 \end{aligned}$$

$$(x, y) \rightarrow (2x-4, y-2)$$

$$(-2, -\frac{1}{2}) \rightarrow (-8, -2.5)$$

$$(-1, -1) \rightarrow (-6, -3)$$

$$(-\frac{1}{2}, -2) \rightarrow (-5, -4)$$

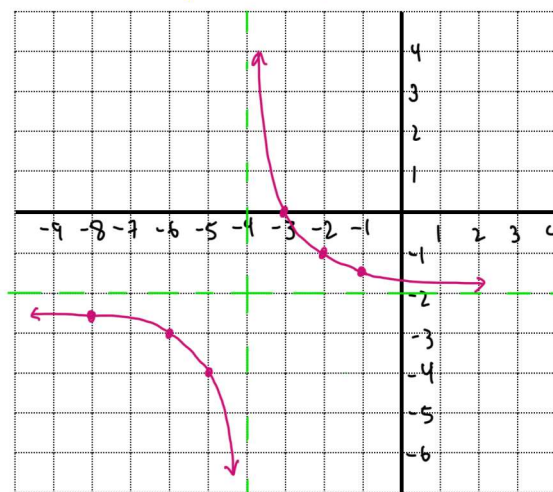
$$(\frac{1}{2}, 2) \rightarrow (-3, 0)$$

$$(1, 1) \rightarrow (-2, -1)$$

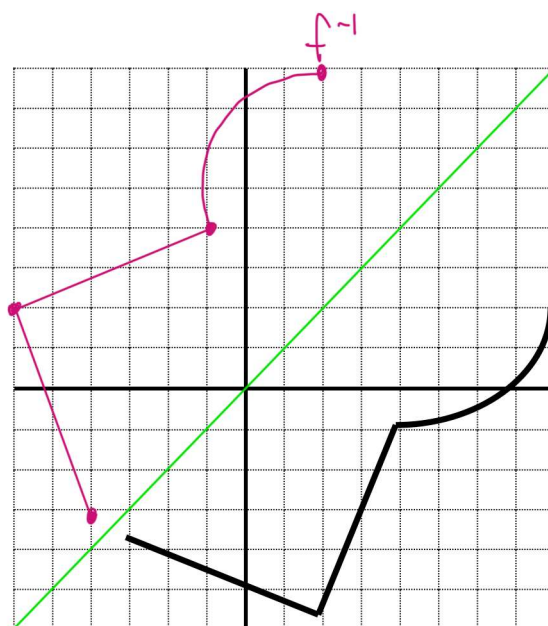
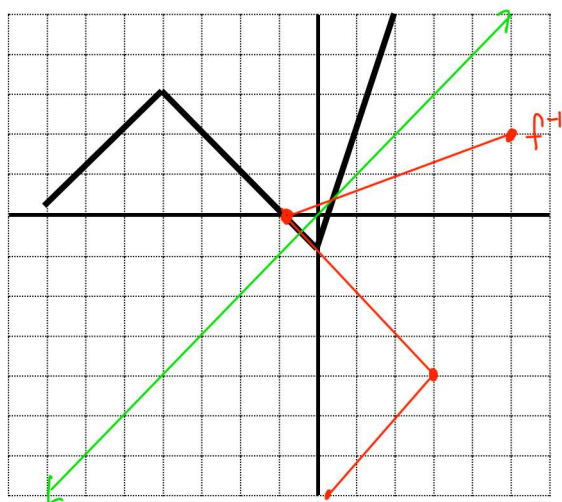
$$(2, \frac{1}{2}) \rightarrow (0, -1.5)$$

$$\text{V.A at } x=0 \rightarrow \text{V.A at } x=-4$$

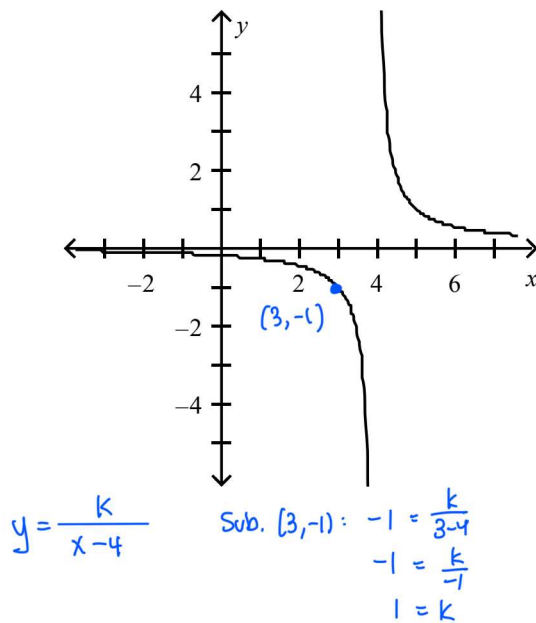
$$\text{H.A at } y=0 \rightarrow \text{H.A at } y=-2$$



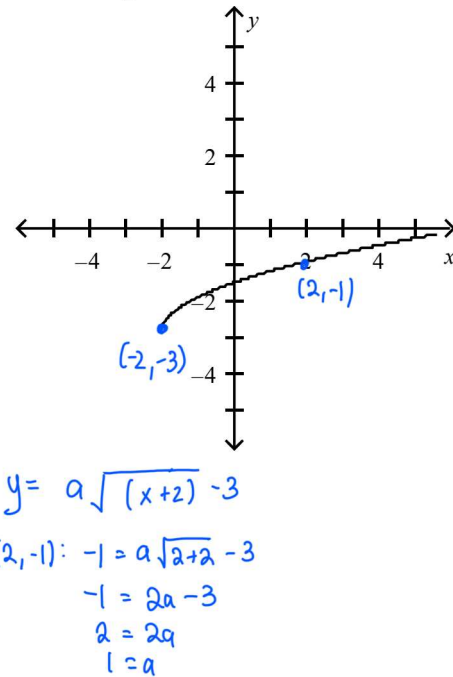
2. Graph the inverse function. *Reflected on $y=x$*



3. Find the equation of the function $f(x)$. 4. Find the equation of the function $f(x)$.



Equation: $f(x) = \frac{1}{x-4}$



Equation: $f(x) = \sqrt{x+2} - 3$

5. Describe the transformations that must be applied to the following:

$f(x) = \frac{1}{x}$ to obtain the graph of $y = -\frac{2}{10x+5} - 7 = \frac{-2}{10(x+\frac{1}{2})} - 7$

- Reflection on x-axis
- Vertical stretch by a factor of 2
- horizontal compression by a factor of $\frac{1}{10}$
- $\frac{1}{2}$ unit left
- 7 units down

6. Find the equation of the transformed functions as described below.

The graph of $y = f(x)$ has been reflected in the x-axis, horizontally compressed by a factor of one-third, shifted up two units. What is the equation of this transformed function?

Equation: $y = -f(3x) + 2$

The graph of $y = \sqrt{x}$ has been reflected in the y-axis, horizontally stretched by a factor of two, shifted left three units. What is the equation of this transformed function?

Equation: $y = \sqrt{-\frac{1}{2}(x+3)}$

7. A function $f(x) = \{(-1, -5), (0, -1), (1, 2), (3, 10)\}$. If the function is transformed into $y = 3f(-x + 2) - 1$, state the transformed points.

$$y = 3f[-(x-2)] - 1$$

$$(x, y) \rightarrow (-x+2, 3y-1)$$

$$\therefore y = \{(3, -16), (2, -4), (1, 5), (-1, 29)\}$$

8. A function $g(x) = \{(-4, -7), (-2, -3), (1, 2), (4, 6)\}$. If the function is transformed into $y = -\frac{1}{2}g(2x - 3) + 1$, state the transformed points.

$$y = -\frac{1}{2}g(2(x-\frac{3}{2})) + 1$$

$$(x, y) \rightarrow (\frac{1}{2}x + \frac{3}{2}, -\frac{1}{2}y + 1)$$

$$\therefore y = \{(-\frac{1}{2}, \frac{9}{2}), (\frac{1}{2}, \frac{5}{2}), (2, 0), (\frac{7}{2}, -2)\}$$

9. A function $f(x) = \{(-1, -5), (0, -1), (1, 2), (3, 10)\}$ is transformed into $g(x) = af(k(x-d)) + c$ where $g(x) = \{(1, 6), (3, -2), (5, -8), (9, -24)\}$. Find the equation of $g(x)$.

$$(x, y) \rightarrow (\frac{1}{k}x + d, ay + c)$$

$$(0, -1) \rightarrow (3, -2)$$

$$\begin{aligned} \hookrightarrow \frac{1}{k}(0) + d &= 3 \\ \therefore d &= 3 \\ \hookrightarrow -a + c &= -2 \quad \text{--- (1)} \end{aligned}$$

$$(-1, -5) \rightarrow (1, 6)$$

$$\begin{aligned} \hookrightarrow \frac{1}{k}(-1) + 3 &= 1 \\ -\frac{1}{k} &= -2 \\ \frac{1}{2} &= k \\ \hookrightarrow -5a + c &= 6 \quad \text{--- (2)} \end{aligned}$$

$$\textcircled{1} - \textcircled{2} : -a + c = -2$$

$$- \quad -5a + c = 6$$

$$4a = -8$$

$$a = -2$$

$$\text{Sub. } a = -2 \text{ into } \textcircled{1} :$$

$$2 + c = -2$$

$$c = -4$$

$$\therefore g(x) = -2f\left(\frac{1}{2}(x-3)\right) - 4$$

10. A function $g(x) = \{(-4, -7), (-2, -3), (1, 2), (4, 6)\}$ is transformed into $g(x) = af(k(x-d)) + c$ where $g(x) = \{(9, -16), (5, -4), (-1, 11), (-7, 23)\}$. Find the equation of $g(x)$.

$$(x, y) \rightarrow (\frac{1}{k}x + d, ay + c)$$

$$(-2, -3) \rightarrow (5, -4)$$

$$\begin{aligned} \hookrightarrow -\frac{3}{k} + d &= 5 \quad \text{--- (1)} \\ -3a + c &= -4 \quad \text{--- (2)} \end{aligned}$$

$$(1, 2) \rightarrow (-1, 11)$$

$$\begin{aligned} \hookrightarrow \frac{1}{k} + d &= -1 \quad \text{--- (3)} \\ 2a + c &= 11 \quad \text{--- (4)} \end{aligned}$$

$$\textcircled{1} - \textcircled{3} : -\frac{3}{k} + d = 5$$

$$- \quad \frac{1}{k} + d = -1$$

$$-\frac{3}{k} = 6$$

$$-\frac{1}{2} = k$$

$$\text{Sub } k = -\frac{1}{2} \text{ into } \textcircled{3} : -2 + d = -1$$

$$d = 1$$

$$\textcircled{2} - \textcircled{4} : -3a + c = -4$$

$$- \quad 2a + c = 11$$

$$-5a = -15$$

$$a = 3$$

$$\text{Sub } a = 3 \text{ into } \textcircled{2} : 6 + c = -4$$

$$c = -10$$

$$\therefore g(x) = 3f\left(-\frac{1}{2}(x-1)\right) - 10$$

11. For a function of the form $y = af(k(x-d)) + c$, describe how the domain and range changes from the base graphs of

a) $f(x) = x^2$ domain does not change. Range: If $a > 0 \rightarrow \{y \in \mathbb{R} \mid y \geq c\}$

b) $f(x) = \frac{1}{x}$ domain: $\{x \in \mathbb{R} \mid x \neq d\}$ Range: $\{y \in \mathbb{R} \mid y \neq c\}$

c) $f(x) = \sqrt{x}$ Domain: If $k > 0 \rightarrow \{x \in \mathbb{R} \mid x \geq d\}$ Range: If $a > 0 \rightarrow \{y \in \mathbb{R} \mid y \geq c\}$
If $k < 0 \rightarrow \{x \in \mathbb{R} \mid x \leq d\}$ If $a < 0 \rightarrow \{y \in \mathbb{R} \mid y \leq c\}$

12. Explain the relationship between a horizontal stretch $y = f(kx)$ and a vertical stretch $y = af(x)$. How are a and k related?

For some functions $\Rightarrow$ A vertical stretch is equivalent to a horizontal compression (and vice versa)
A vertical compression is equivalent to a horizontal stretch (and vice versa)